

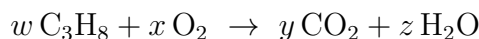
NATIONAL UNIVERSITY OF SINGAPORE
Department of Mathematics

MA1513
Additional Practice Set 1

1. Write down the augmented matrix of the following linear system. Reduce the matrix to row echelon form. Hence find the solutions of the system.

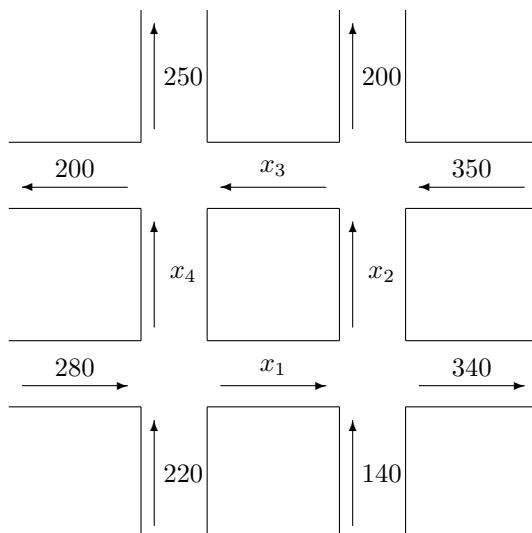
$$\begin{array}{rrcr} x & + & y & + & 2z & = & 8 \\ -x & - & 2y & + & 3z & = & 1 \\ 3x & - & 7y & + & 4z & = & 10 \end{array}$$

2. The equation $ax^2 + ay^2 + bx + cy + d = 0$ represents a curve on the xy -plane that passes through three points $(-4, 5)$, $(-2, 7)$ and $(4, -3)$. Use Gaussian elimination to find a, b, c and d .
3. When propane gas burns, the propane combines with oxygen to form carbon dioxide and water:



where w and x are the numbers of propane and oxygen molecules, respectively, required for the combustion; and y and z are the numbers of carbon dioxide and water molecules, respectively, produced.

- (a) By equating the numbers of carbon, hydrogen and oxygen atoms, respectively, on both sides of the chemical equation, write down a homogeneous system of three equations in terms of w, x, y, z .
- (b) Find a general solution for the homogeneous system obtained in Part (a).
- (c) Find the (non-trivial) solution of w, x, y, z with smallest values.
4. In a shopping district in the western part of Singapore, two parallel pairs of one-way streets intersect as shown below:



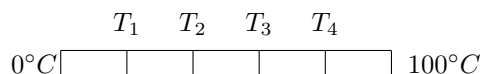
The average hourly volume of traffic entering and leaving this section during peak hour is given in the diagram.

- (a) Set up a linear system with four equations and four variables x_1, x_2, x_3, x_4 . (You do not need to solve the system.)
- (b) Given the general solution for the linear system is

$$x_1 = 500 - t, x_2 = 300 - t, x_3 = 450 - t, x_4 = t,$$

find the range for the possible values of t .

- (c) Given that $x_1 = x_4$, find x_1, x_2, x_3 and x_4 .
5. An insulated metal rod has temperature 0 and 100 degree Celsius at the two ends as shown in the diagram:



T_1, T_2, T_3, T_4 represent the temperatures at the indicated nodes. Assume that the temperature at each node is the average of the temperature at the two neighbouring nodes.

- (i) Set up a linear system with four equations and four variables T_1, T_2, T_3, T_4 which could be used to determine the temperatures at the nodes.
 - (ii) Express the system in (i) in matrix equation form $\mathbf{Ax} = \mathbf{b}$.
 - (iii) Show that $\mathbf{x} = \begin{pmatrix} 20 \\ 40 \\ 60 \\ 80 \end{pmatrix}$ is a solution of the system in (i).
 - (iv) Does the system in (i) have any other solution?
6. Let $\mathbf{B}$ be a 3×5 matrix and $\mathbf{d}$ a 3×1 column matrix. Suppose the non-homogeneous linear system $\mathbf{Bx} = \mathbf{d}$ has a solution

$$x_1 = 1, \quad x_2 = 1, \quad x_3 = -1, \quad x_4 = 2, \quad x_5 = 1$$

and the homogeneous system $\mathbf{Bx} = \mathbf{0}$ has a general solution

$$x_1 = t - s, \quad x_2 = s + 2t, \quad x_3 = 0, \quad x_4 = s, \quad x_5 = t$$

where s and t are free parameters.

- (i) Find the general solution of $\mathbf{Bx} = \mathbf{d}$.
- (ii) Write down the reduced row echelon form of $\mathbf{B}$.

7. Given a 3×5 matrix $\mathbf{A}$ with row echelon form:

$$\mathbf{R} = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

and a 3×1 non-zero vector $\mathbf{b}$.

- (i) Write down a general solution of the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

- (ii) Given $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$ is a solution of the non-homogeneous system $\mathbf{Ax} = \mathbf{b}$. Write down a general solution of this system.

- (iii) If $\mathbf{C}$ is another 3×5 matrix with the same row echelon form $\mathbf{R}$, do we have enough information to find the general solutions of $\mathbf{Cx} = \mathbf{0}$ and $\mathbf{Cx} = \mathbf{b}$?

8. Let $\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ and

$$\mathbf{E}_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}.$$

- (a) Pre-multiply each of $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$ to $\mathbf{A}$.
- (b) Each of the multiplications in part (a) has the same effect on $\mathbf{A}$ as performing an elementary row operation. Identify the elementary row operation that corresponds to each of $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3$.
9. (a) If $\mathbf{A}$ is a random square matrix, is it always true that $\mathbf{A} + \mathbf{A}^T$ is a symmetric matrix?
- (b) If $\mathbf{A}$ is a random matrix (not necessarily square), is it always true that $\mathbf{AA}^T$ is a symmetric matrix?
10. Consider a population model for a certain species of animals that has two life stages: juvenile (up to 1 year old) and adult. Each year,
- an adult produces an average of 1.6 juveniles;
 - 30% of the juveniles survive to become adults;
 - 80% of the adults survive.

Let $\mathbf{x}_n = \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .

- (i) Find the stage matrix $\mathbf{A}$ such that $\mathbf{x}_{n+1} = \mathbf{Ax}_n$.
- (ii) Find the inverse matrix $\mathbf{A}^{-1}$.
- (iii) Suppose the number of juveniles and adults are 480 and 360 respectively in year 2019. Find $j_{2018}, a_{2018}, j_{2020}, a_{2020}$.